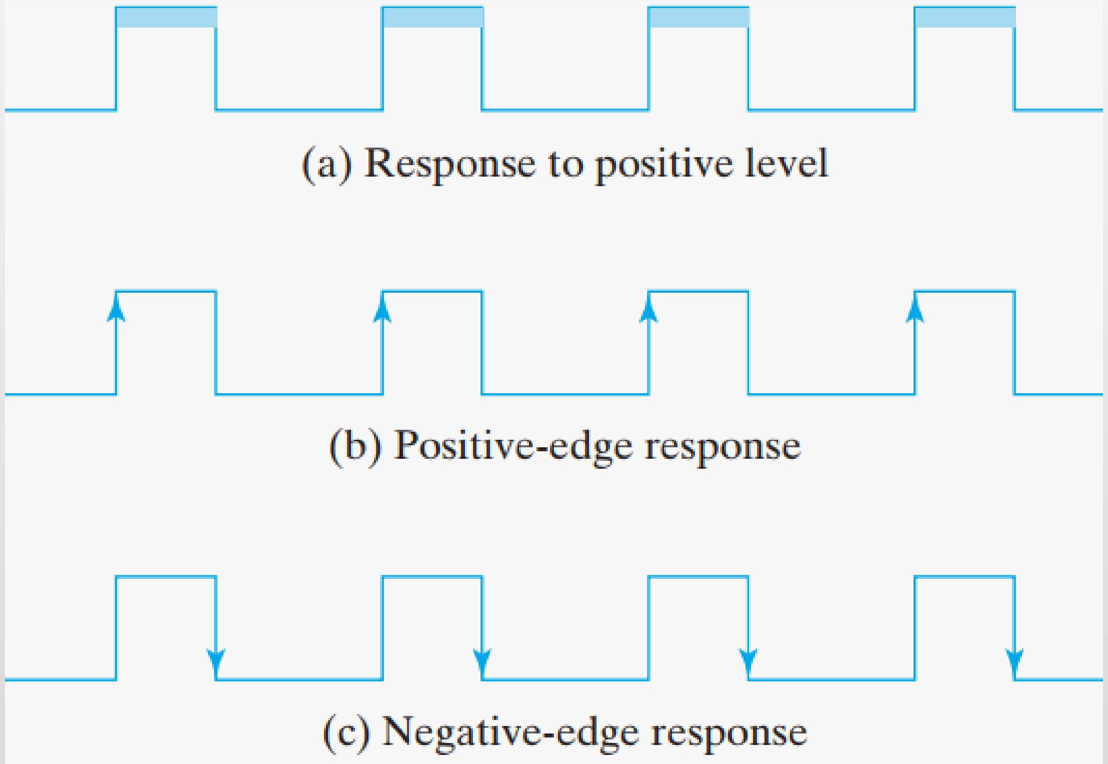
PART 4: SEQUENTIAL LOGIC

LATCHES & FLIP FLOPS

SAAD BAIG – DIGITAL LOGIC DESIGN

STORAGE ELEMENTS

- A storage element in a digital circuit can maintain a binary state indefinitely (as long as power is delivered to the circuit), until directed by an input signal to switch states.
- Storage elements that operate with signal levels (rather than signal transitions) are referred to as latches.
- Storage elements that controlled by a clock transition are flip-flops.



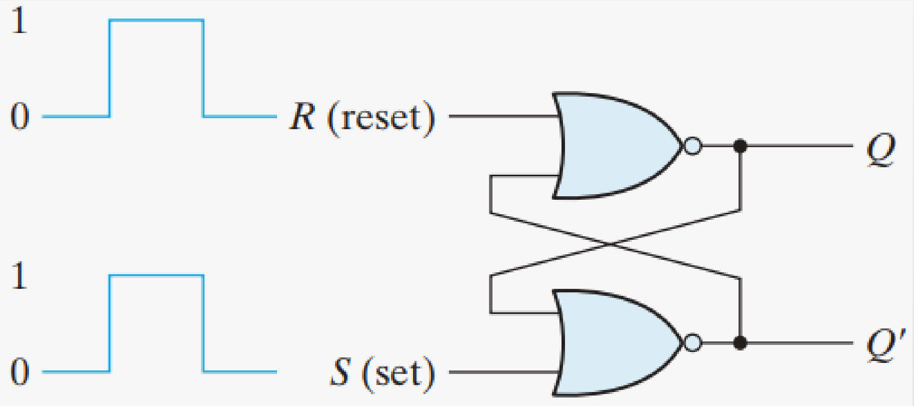
(a) Response to positive level

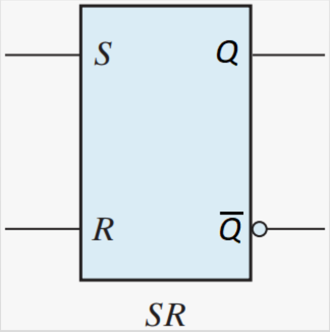
(b) Positive-edge response

(c) Negative-edge response

SR-LATCH

- Inputs: S, R.
- Outputs: Q, $\overline{\text{Q}}$
- States:
  1. Set State: $\text{Q} = 1, \overline{\text{Q}} = 0$ (when S = 1, R = 0)
  2. Reset State: $\text{Q} = 0, \overline{\text{Q}} = 1$ (when S = 0, R = 1)
  - Forbidden State: $\text{Q} = 0, \overline{\text{Q}} = 0$ (when S = 1, R = 1)

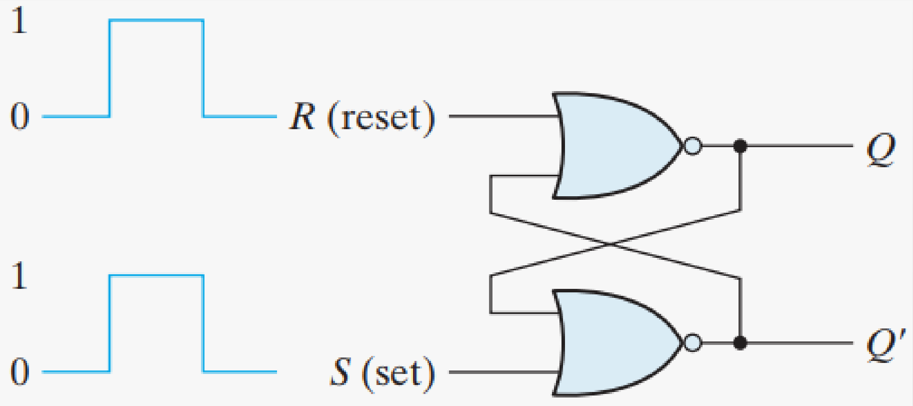


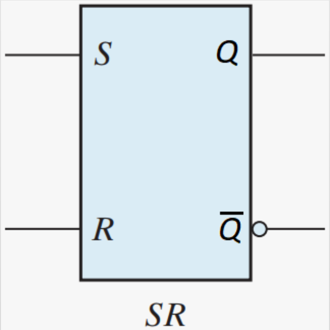


S	R	Q	Q'	
1	0	1	0	
0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	(forbidden)

SR-LATCH

- When S = 1, R = 0 $\rightarrow Q_t = 1$, $Q_t' = 0$
  The set state.
- When S = 0, R = 1 $\rightarrow Q_t = 0$, $Q_t' = 1$
  The reset state.
- When S = 0, R = 0 $\rightarrow Q_t = Q_{t-1}$, $Q_t' = Q_{t-1}'$
  The memory state.
- When S = 1, R = 1 $\rightarrow Q_t = Q'_t = 0$,
  $\rightarrow Q_{t+1} = Q'_{t+1}$ = undefined
  The forbidden state.

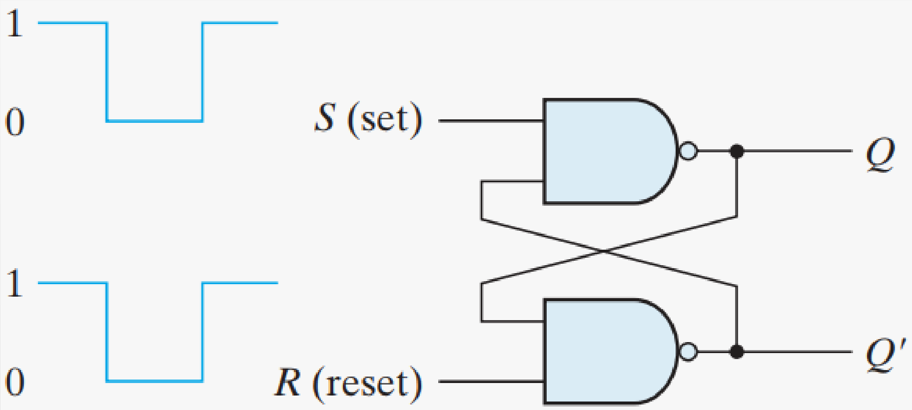


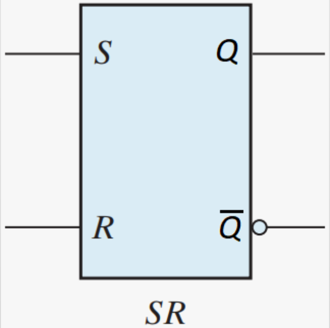


S	R	Q	Q'	
1	0	1	0	
0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	(forbidden)

$\overline{S}\overline{R}$-LATCH

- Inputs: S, R.
- Outputs: Q, $\overline{\text{Q}}$
- States:
  1. Set State: $\text{Q} = 1, \overline{\text{Q}} = 0$ (when S = 0, R = 1)
  2. Reset State: $\text{Q} = 0, \overline{\text{Q}} = 1$ (when S = 1, R = 0)
  - Forbidden State: $\text{Q} = 1, \overline{\text{Q}} = 1$ (when S = 0, R = 0)

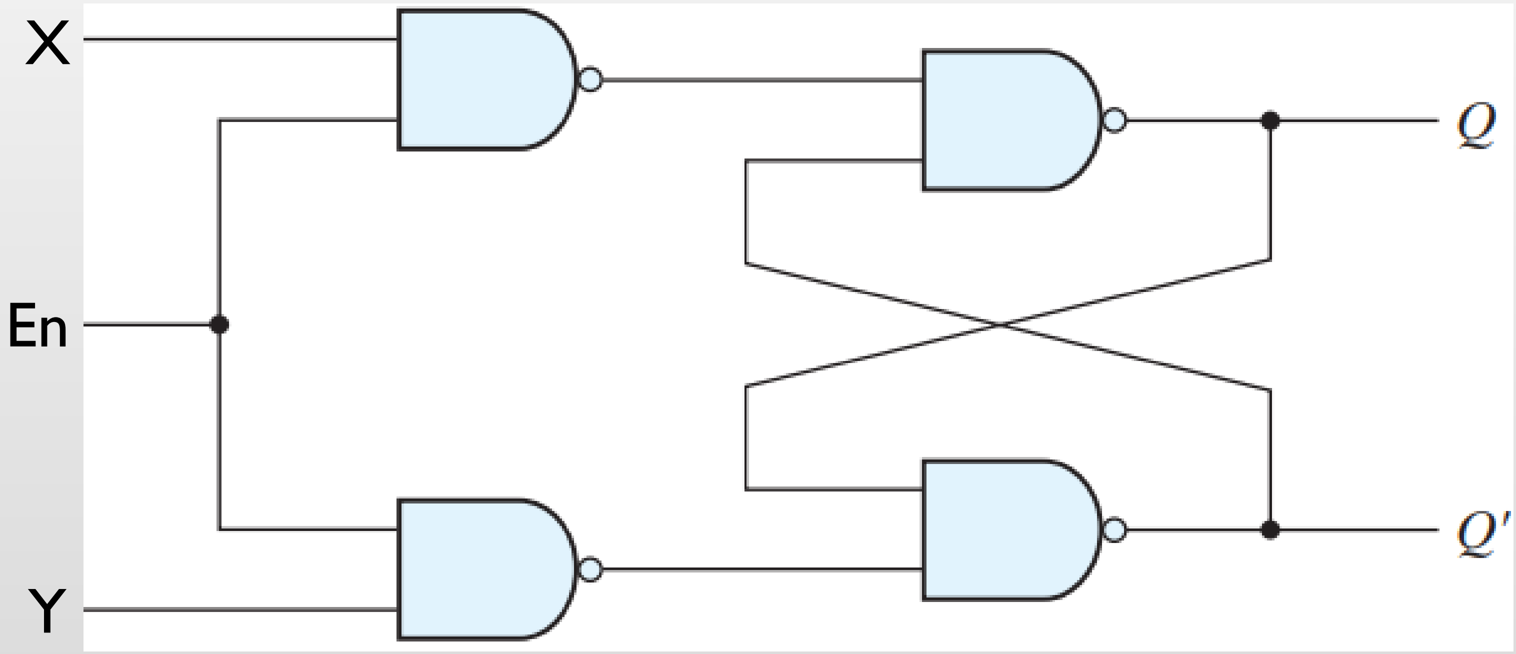




S	R	Q	Q'	
1	0	0	1	
1	1	0	1	(after $S = 1, R = 0$)
0	1	1	0	
1	1	1	0	(after $S = 0, R = 1$)
0	0	1	1	(forbidden)

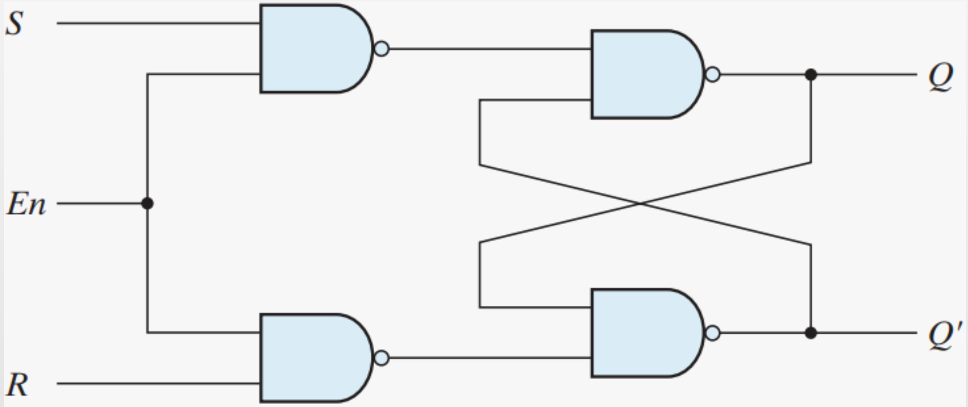
ACTIVITY

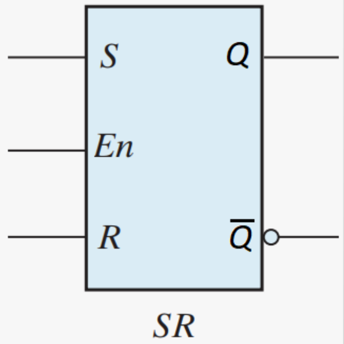
Determine the function table of below latch circuit.



GATED SR-LATCH

- Inputs: S, R, En
- Outputs: $Q, \overline{Q}$
- States (En = 1):
  1. Set State: $Q = 1, \overline{Q} = 0$ (when S = 1, R = 0)
  2. Reset State: $Q = 0, \overline{Q} = 1$ (when S = 0, R = 1)
  - Forbidden State: $Q = 1, \overline{Q} = 1$ (when S = 1, R = 1)

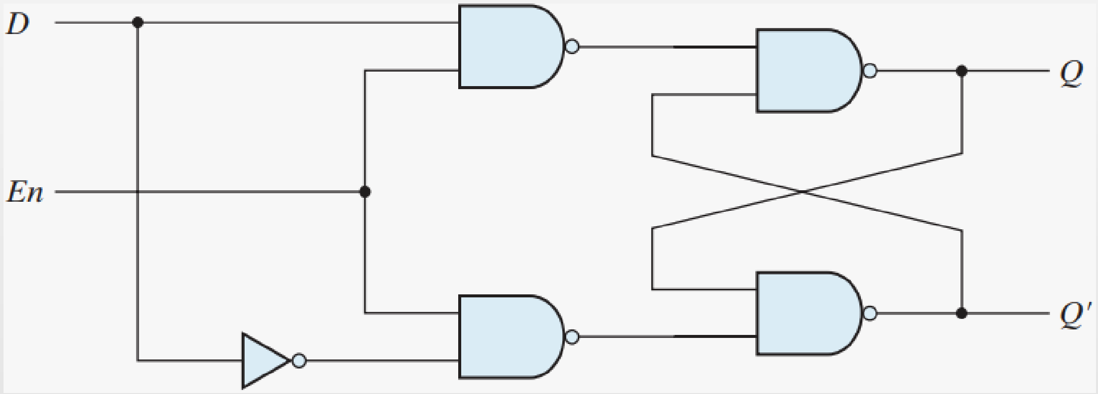


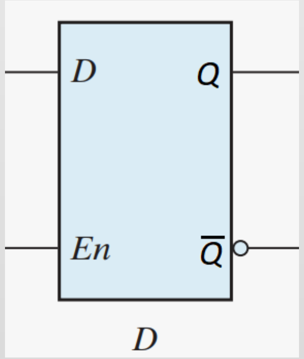


En	*S*	*R*	Next state of *Q*
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

D-LATCH

- Inputs: D, En
- Outputs: $Q, \overline{Q}$
- States (En = 1):
    1. Set State: $Q = 1, \overline{Q} = 0$ (when D = 1)
    2. Reset State: $Q = 0, \overline{Q} = 1$ (when D = 0)
- Now the undeterminable state no longer exists!





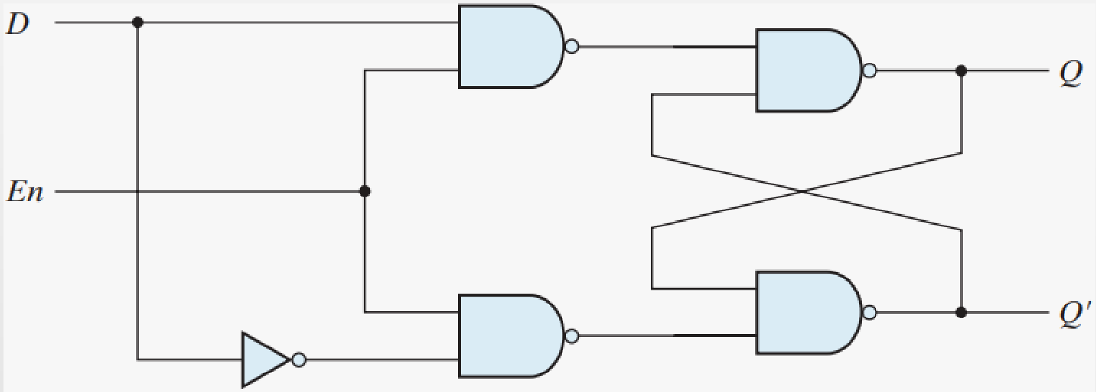
En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

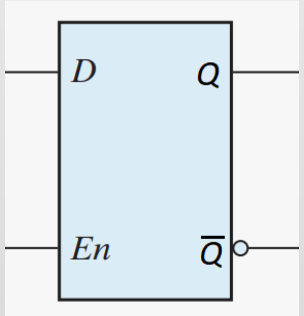
ISSUES WITH LATCHES

- A major problem with the latch: It responds to a change in the *level* of a clock pulse
- If the inputs applied to the latches change while the clock pulse is still at the logic-1 level, the latches will respond to new values and a new output state may occur, causing unpredictability.
- Hence the output of a latch cannot be applied directly or through combinational logic to the input of the same or another latch when all the latches are triggered by a common clock source.
- The key to the proper operation of a flip-flop is to trigger it only during a signal *transition.* That's where flip flops shine.

D-LATCH

- Inputs: D, En
- Outputs: Q, $\overline{\mathrm{Q}}$
- States (En = 1):
  1. Set State: $\mathrm{Q} = 1, \overline{\mathrm{Q}} = 0$ (when D = 1)
  2. Reset State: $\mathrm{Q} = 0, \overline{\mathrm{Q}} = 1$ (when D = 0)





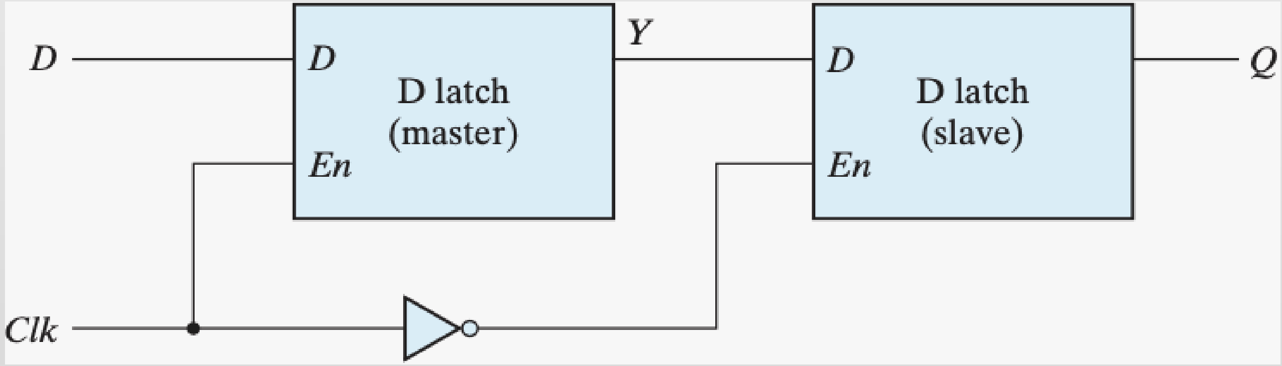
D

En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

D-FLIP FLOP

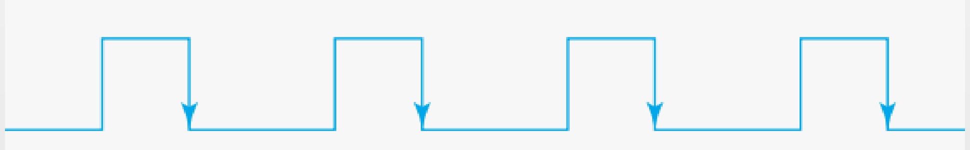
- D-FF constructed with two D-latches – master and slave
- Clk is high → the master D-latch samples the D signal; Y=D.
- The slave D-latch is disabled; Q retains the previous value.
- Clk is low → the master D-latch is disabled; Y= retain its previous value.
- The slave D-latch is enabled; Q is updated with a value of Y

En	D	Next state of Q
0	X	No change
1	0	Q = 0; reset state
1	1	Q = 1; set state

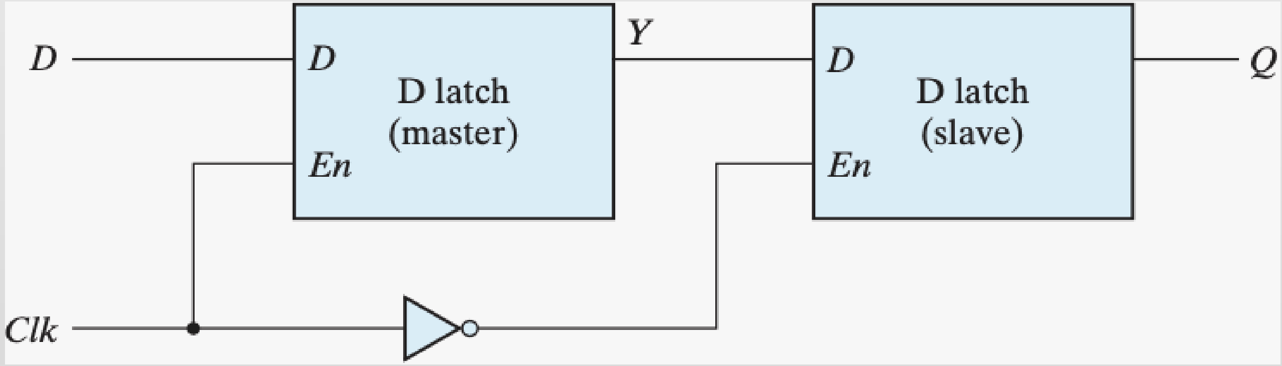


D-FLIP FLOP

- Which edge is this flip flop triggered by?
- This is a negative edge triggered flip flop.
- The output may change only once.

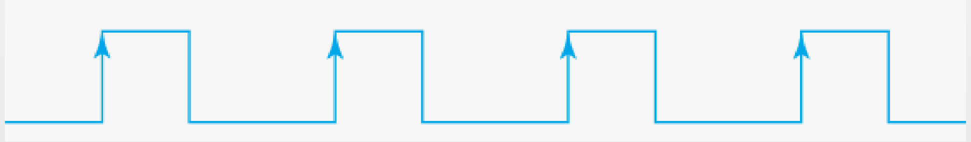


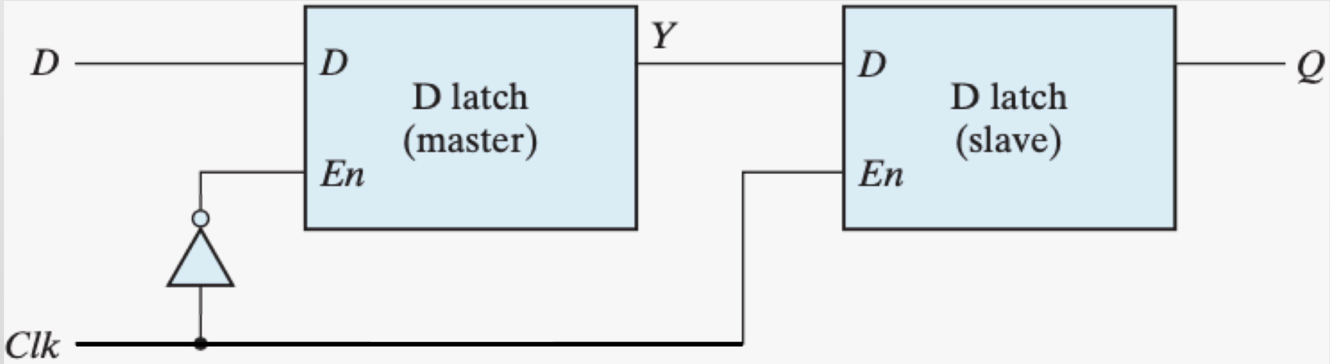
En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state



D-FLIP FLOP

- How can we make it positive edge triggered?

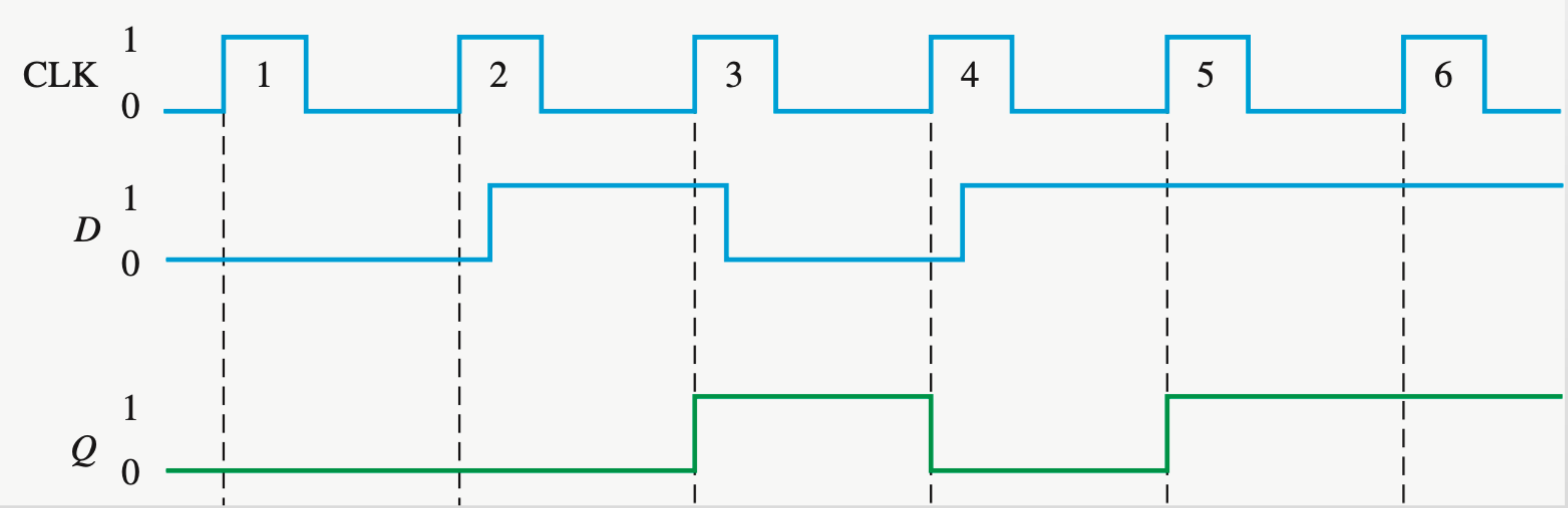




En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

ACTIVITY

- Determine the waveform of Q for the D flip-flop. The input pattern of D and Clk signals are shown below. Assume that D-FF is initially in the RESET state.

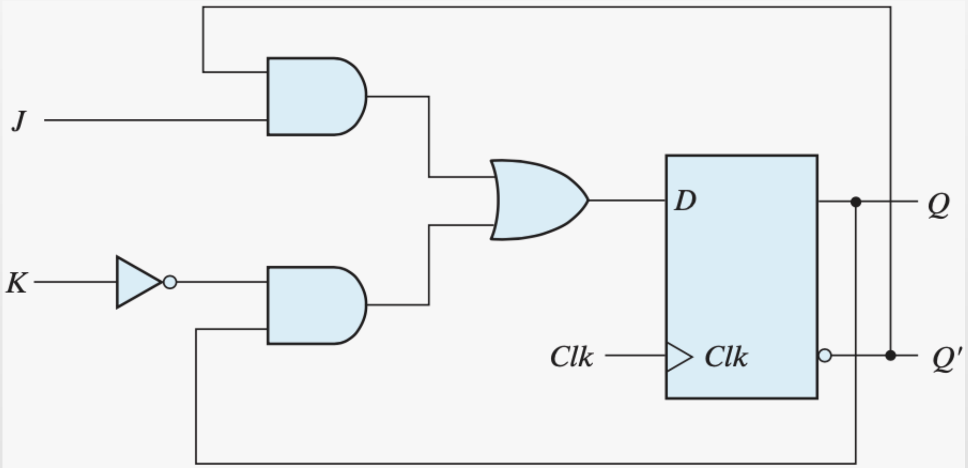


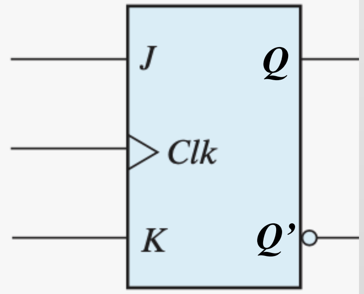
FLIP FLOPS

- Three main operations that can be performed with a flip-flop are:
  - Set the output to 1.
  - Reset the output to 0.
  - Complement the output.
- The most economical and efficient flip-flop is the edge-triggered D flip- flop.
- Other types of flip-flops can be constructed by using the D flip-flop and external logic – JK and T flip-flops.

JK-FLIP FLOP

- D = JQ’ +K’Q
- J = 0, K = 0 → D = Q(t), Q(t+1) = Q(t)
- J = 0, K = 1 → D = 0, Q(t+1) = 0 → Reset state
- J = 1, K = 0 → D = 1, Q(t+1) = 1 → Set state
- J = 1, K = 1 → D = Q’(t), Q(t+1) = Q(t) → Toggle





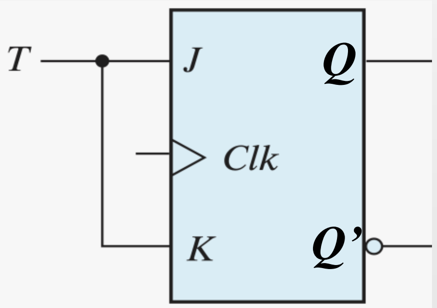
***JK* Flip-Flop**

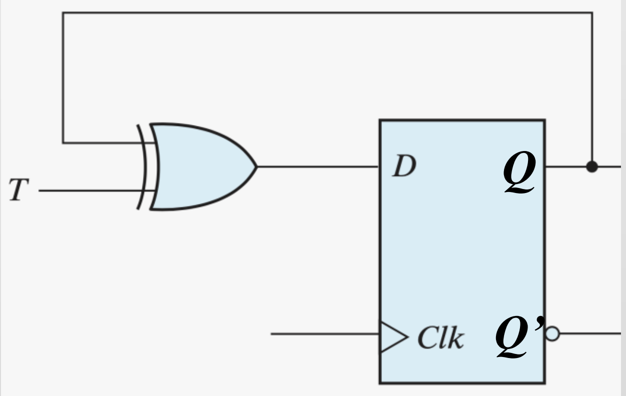
J	K	Q(t + 1)	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

T FLIP FLOP

- Toggle (T) flip-flop is a complementing flip-flop obtained:

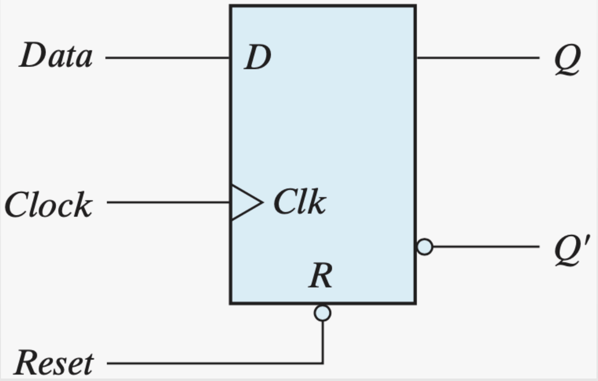
1. From JK flip-flop with J and K inputs tied together.
   - When T = 0 (J = K = 0) → No change
   - When T = 1 (J = K = 1) → Clk edge complements the output
2. From D flip-flop and an XOR gate
   - D = T'Q + TQ'
   - When T = 0 → D = Q
   - When T = 1 → D = Q'





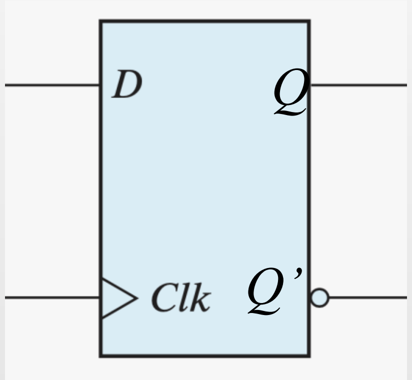
FLIP FLOPS – RESET FUNCTION

- Direct inputs drive the FF to a certain state independently of the clock.
- Direct Set drive the FF to set state (Q = 1 and Q’ = 0).
- Direct Reset or Clear drive the FF to reset state (Q = 0 and Q’ = 1)



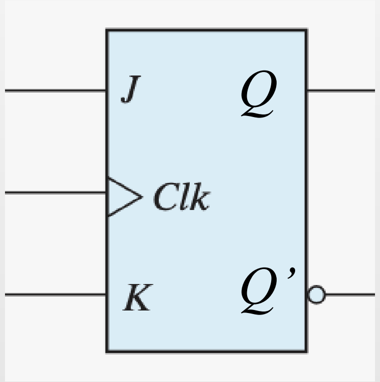
R	Clk	D	Q	Q'
0	X	X	0	1
1	↑	0	0	1
1	↑	1	1	0

FLIP FLOPS SUMMARY



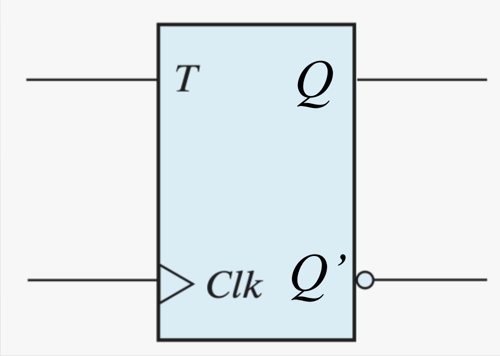
D	$Q(t+1)$	
0	0	Reset
1	1	Set

$Q(t + 1) = D$



J	K	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

$Q(t + 1) = JQ' + K'Q$



T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

$Q(t + 1) = TQ' + T'Q$